

316-221 Metallurgy

Chapter 9

Powder Metallurgy



Overview

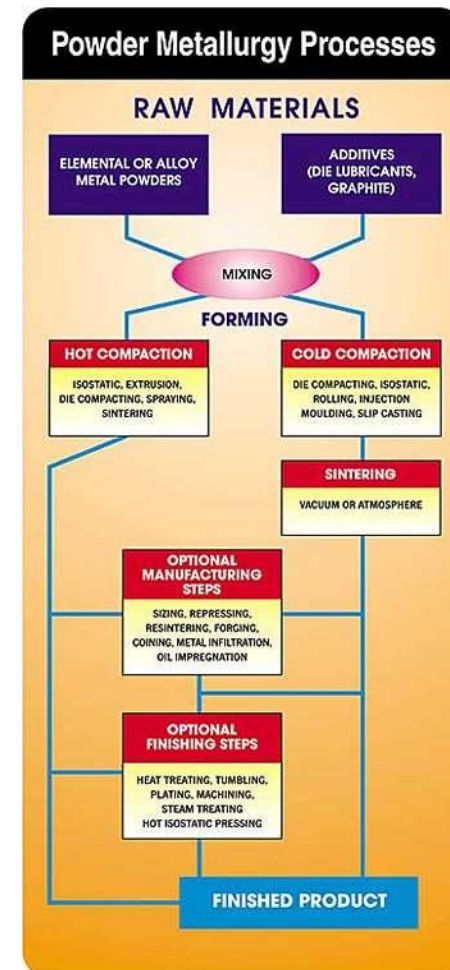
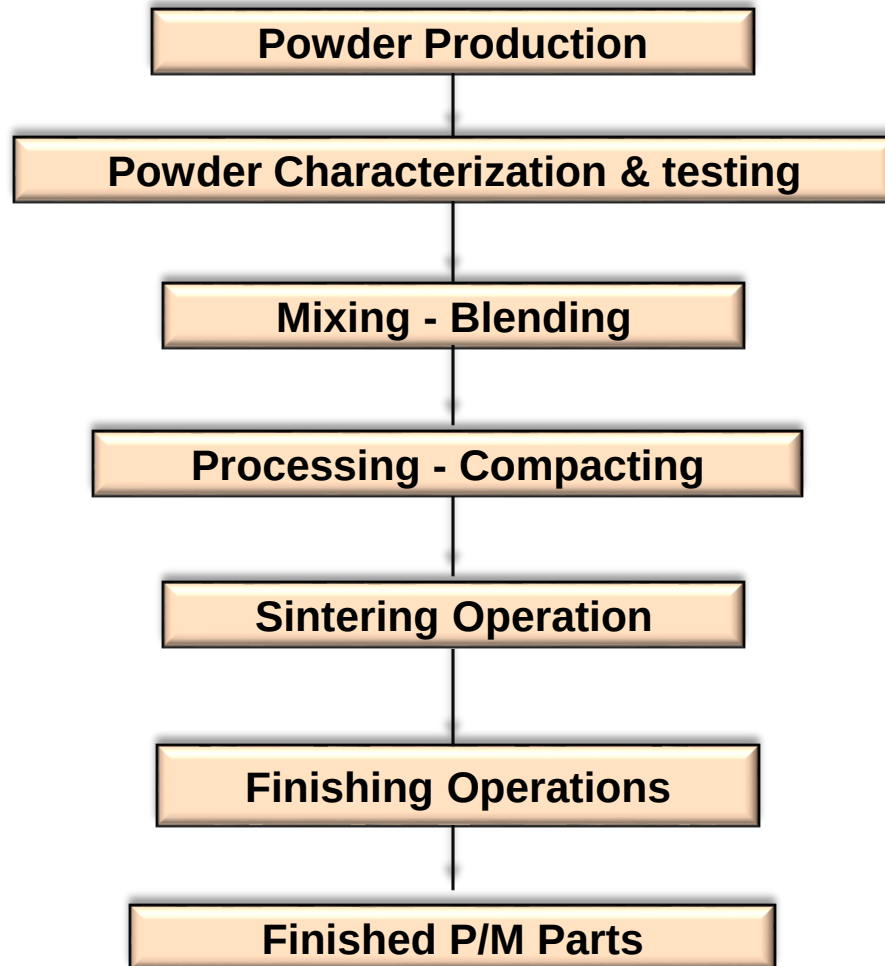
- ▶ Introduction
- ▶ Powder Fabrication
- ▶ Powder Characterization
- ▶ Powder Compaction
- ▶ Powder Sintering
- ▶ Powder Applications

Introduction

- ❑ Powder metallurgy is the name given to the process by which fine powdered materials are **blended**, **pressed** into a desired shape, and then **heated to bond** surfaces
- ❑ Typically used when large amounts of small, intricate parts with high precision are required.
- ❑ Little material waste and unusual mixtures can be utilized.
- ❑ Used for parts in the automotive industry, household appliances, and recreational equipment.

Introduction

The process of P/M in general consists of a series of steps/stages to form a final shape. These stages are shown by a simple flow sheet diagram.



Introduction

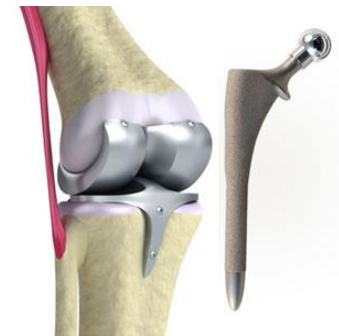
- ❑ Properties of powder metallurgy products are highly dependent on the characteristics of starting powders
- ❑ Some important properties and characteristics
 - Chemistry and purity
 - Particle size
 - Size distribution
 - Particle shape
 - Surface texture
- ❑ Useful in producing prealloyed powders
 - Each powder particle can have the desired alloy composition

Introduction

- **Structural components:** automotive parts

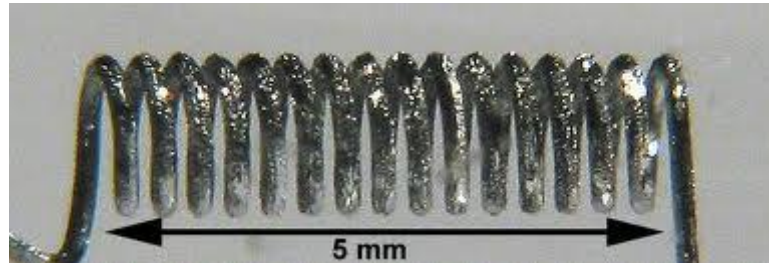


- **Controlled porosity applications:** bearings, filters, sound absorbers, and biomedical implants

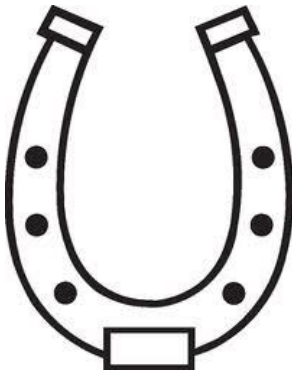


Introduction

- **Electrical applications:** tungsten lamp filament



- **Wear resistance applications:** endurance racing horseshoe



Introduction

Benefits

1. **PM parts can be fabricated to final or near-net shape, thereby eliminating or reducing scrap metal, machining and assembly operation**
2. **Lower overall cost (less scrap lost)**
3. **High melting point metals and composite materials can be produced**
4. **PM is useful in making parts that have complex shapes or difficult to machine**
5. **Alloying elements can be controlled**
6. **Produces good surface finishes**

Introduction

❑ Limitations and Disadvantages

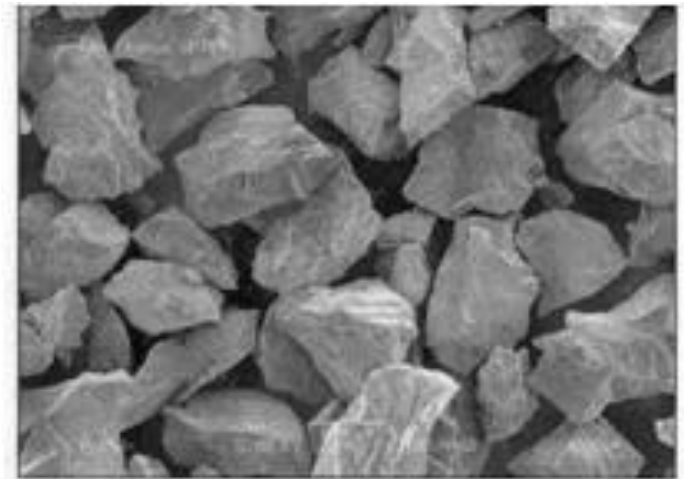
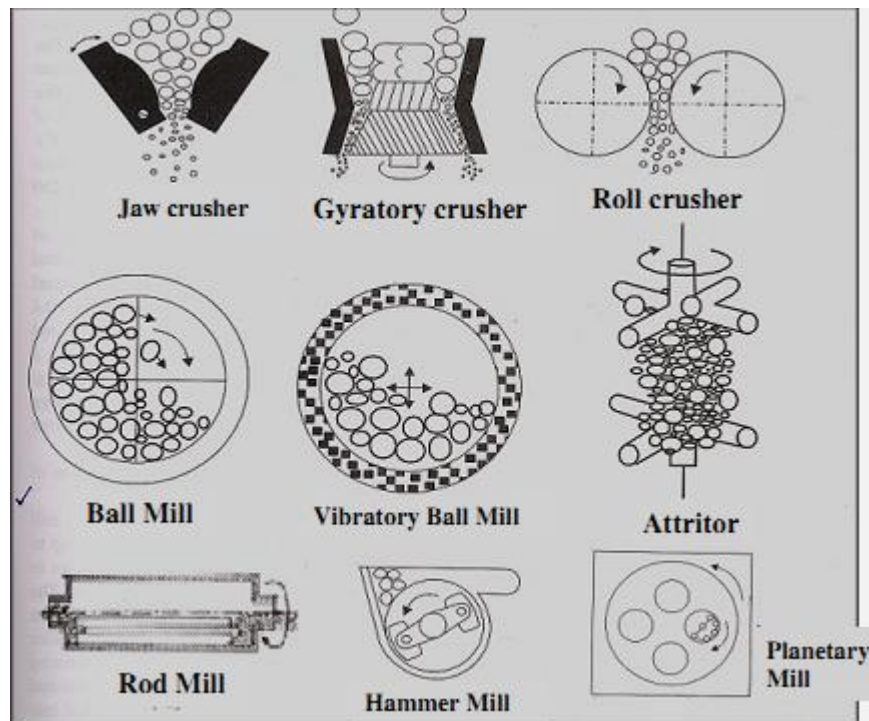
- High tooling and equipment costs
- Metallic powders are expensive
- Problems in storing and handling metal powders
 - Degradation over time, fire hazards with certain metals
- Limitations on part geometry because metal powders do not readily flow well
- Variations in density may be a problem, especially for complex geometries

Powder Fabrication

- **Mechanical Fabrication Techniques**
- **Electrolytic Fabrication Techniques**
- **Chemical Fabrication Techniques**
- **Atomization Techniques**

Mechanical Fabrication Techniques

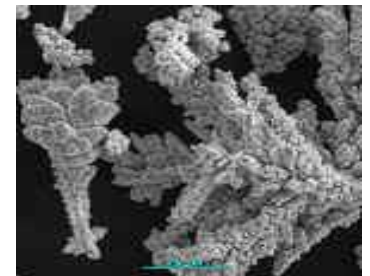
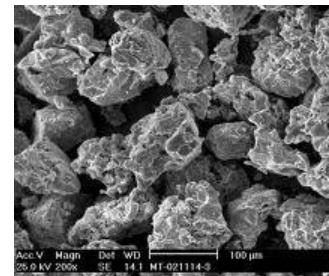
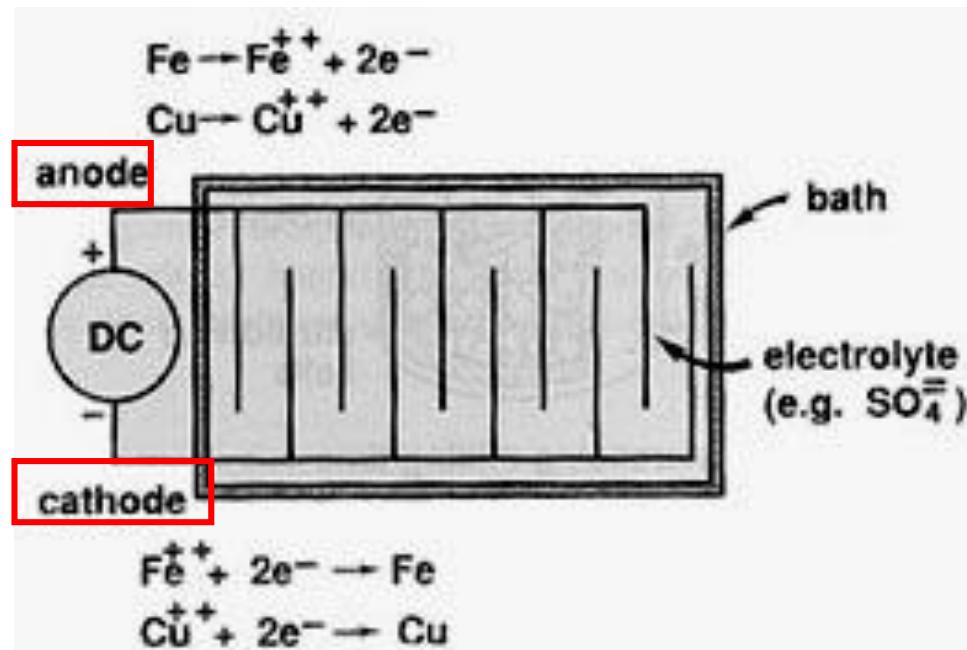
Mechanical Methods: These methods involve using mechanical forces such as compressive forces, shear or impact to facilitate particle size reduction of bulk materials



Angular particle shape

Electrolytic Fabrication Techniques

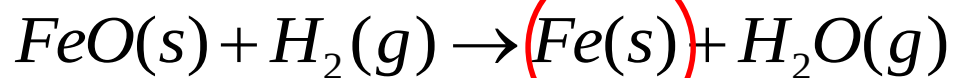
The processing conditions are so chosen that metals of high purity are precipitated from aqueous solution on the cathode of an electrolytic cell. This method is mainly used for producing copper, iron powders. (zinc, tin, nickel, cadmium, antimony, silver, lead, beryllium powders)



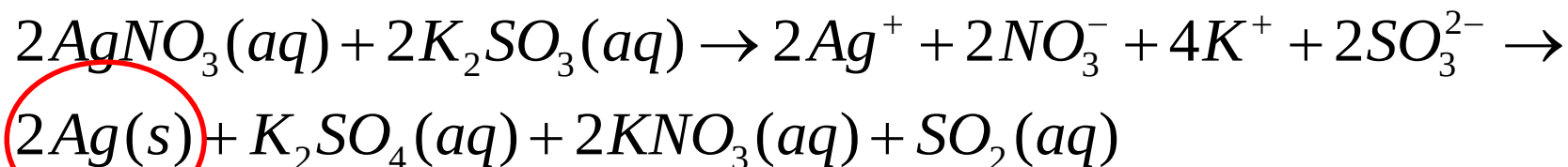
Sponge and dendritic shape

Chemical Fabrication Techniques

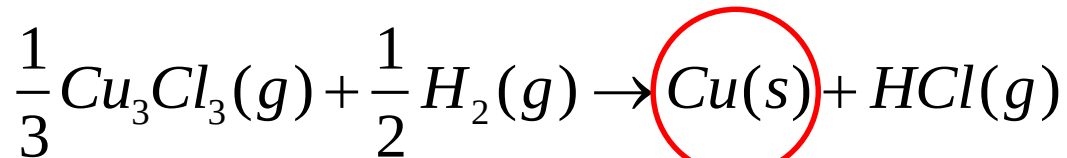
▪ Decomposition of a Solid by a Gas



▪ Precipitation from a Liquid



▪ Precipitation from a Gas

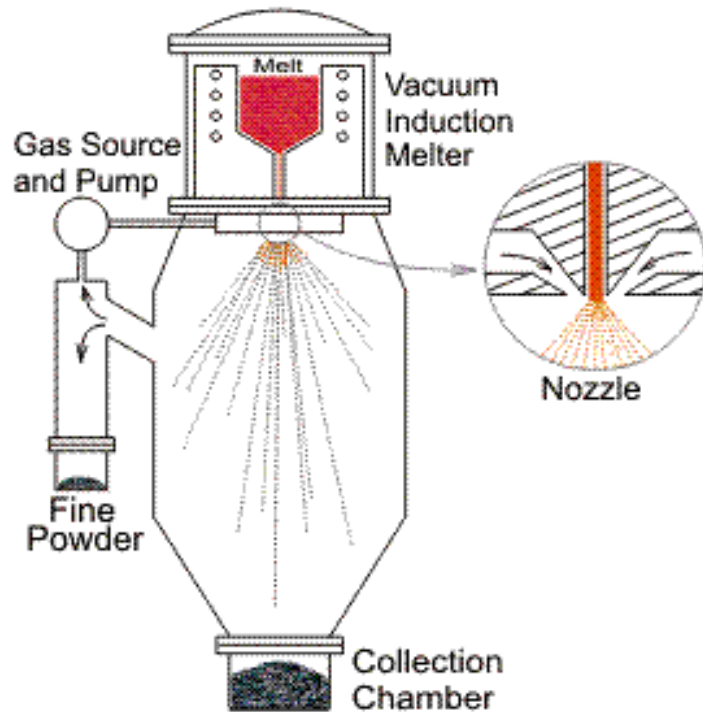


Atomization Techniques

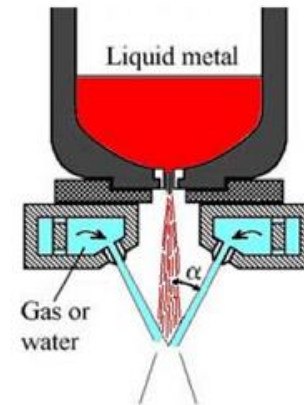
- **Gas Atomization**
- **Water Atomization**
- **Centrifugal Atomization**
- **Ultrasonic Atomization**
- **Hybrid Atomization**

Gas Atomization

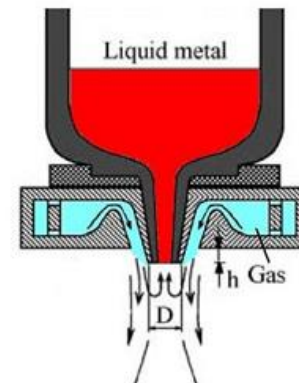
Use of air, nitrogen, helium or argon as a fluid for breaking up a molten metal stream provides an array of versatile powder fabrication techniques



Vertical Gas Atomizer (ref. R.M. German)



Free-fall nozzle



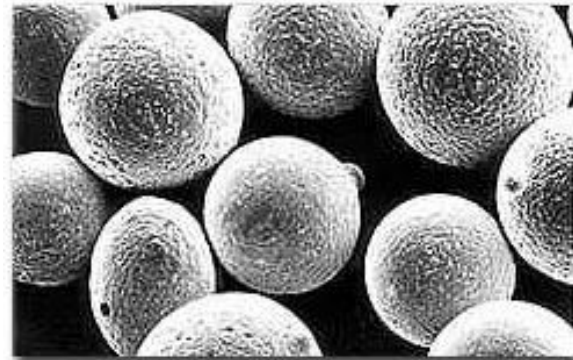
Close-coupled nozzle

Gas Atomization

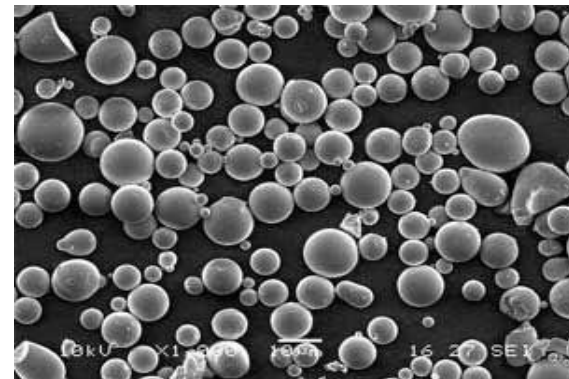
The commercially used for the production of iron, copper, including tool steels, alloy steels, brass, bronze and the low-melting-point metals, such as aluminium, tin, lead, zinc, cadmium.

$$D = \frac{C}{V} \left(\frac{\gamma_m}{\rho_m} \right)^{0.22} \left(\frac{\eta_m}{\rho_m} \right)^{0.57}$$

- D is the particle size
- V is the gas velocity
- C is the nozzle geometry constant
- γ_m is the melt surface energy
- η_m is the melt viscosity
- ρ_m is the melt density



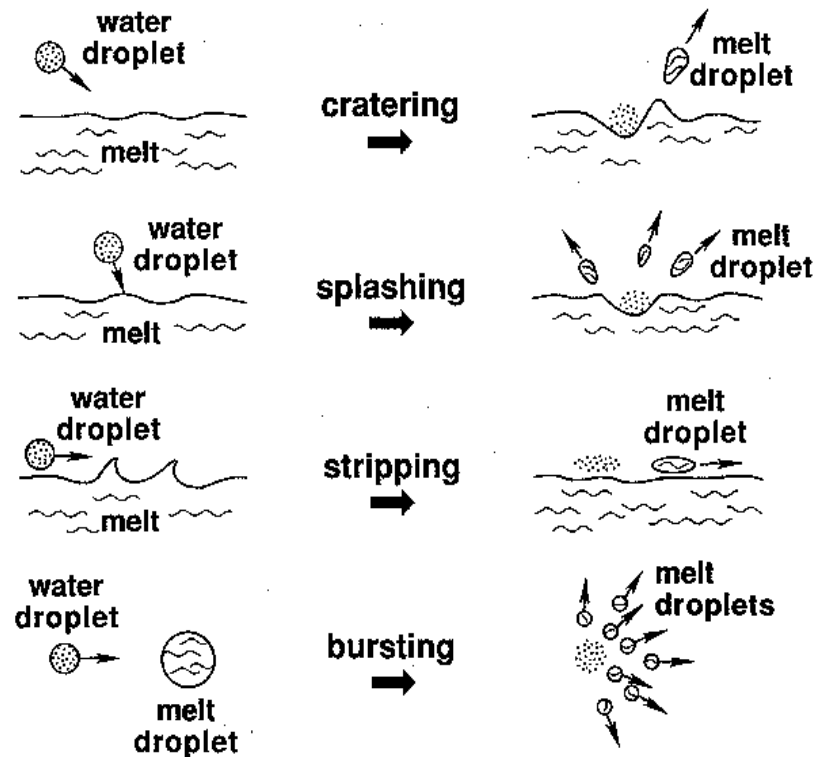
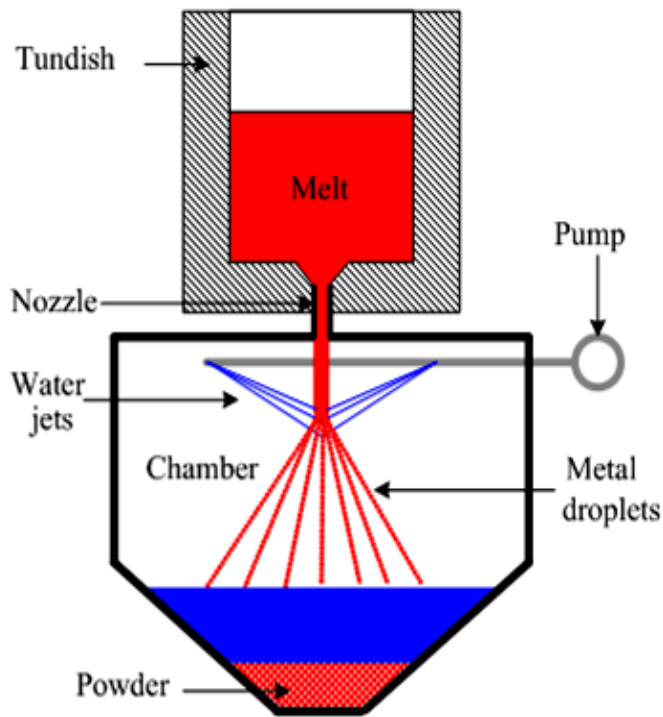
Copper powder



Aluminium powder

Water Atomization

Water atomization is the most common technique for producing elemental and alloy powder from metals which melt below approximately 1600 °C



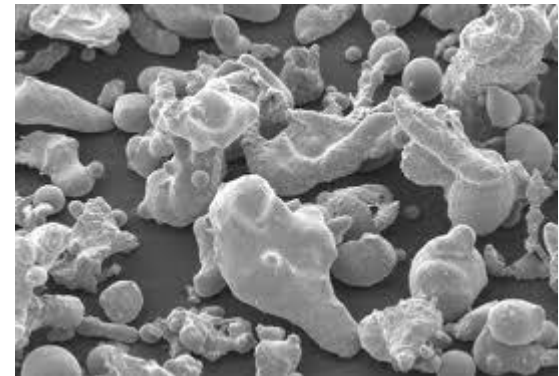
Mechanisms of melt disintegration by WA

Water Atomization

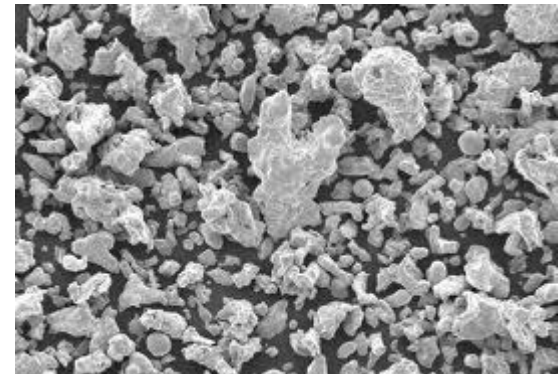
Mathematical models linking the particle size in water atomization. A high water velocity, causes a decrease in the mean particle size. In a simple form, the relation can be expressed as follows:

$$D = \frac{\beta \ln(P)}{V \sin(\alpha)}$$

- D is the particle size
- β is the constant
- P is the atomizer pressure
- V is the water velocity
- α is the angle between melt stream and the water nozzle



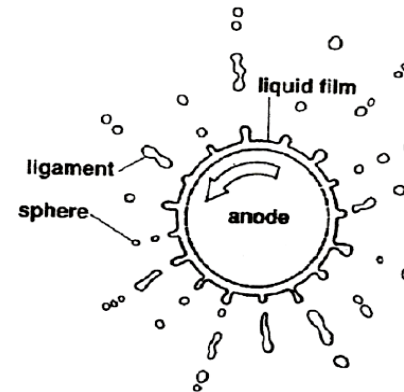
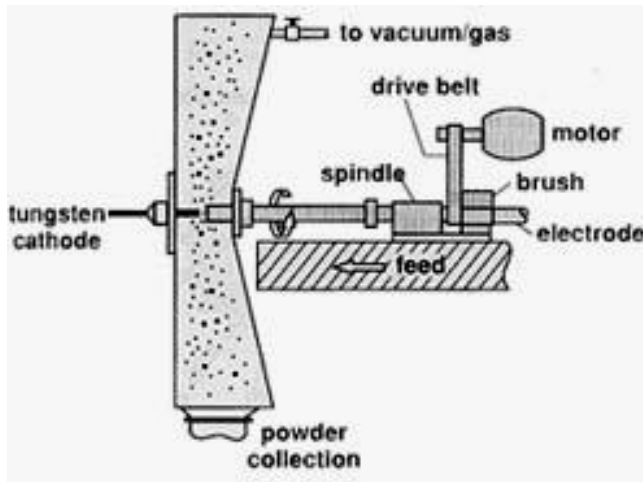
Iron powder



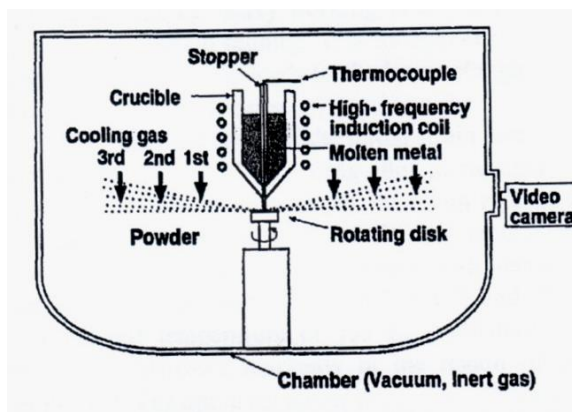
Stainless steel powder

Centrifugal Atomization

Centrifugal atomization is used centrifugal force for melt disintegration to small droplets



Liquid film break-up and formation into small droplets

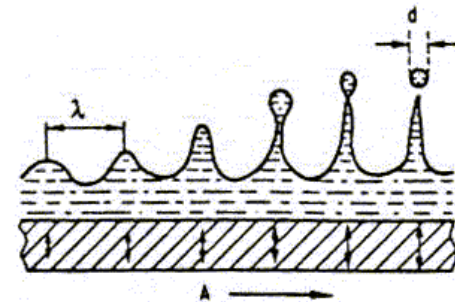
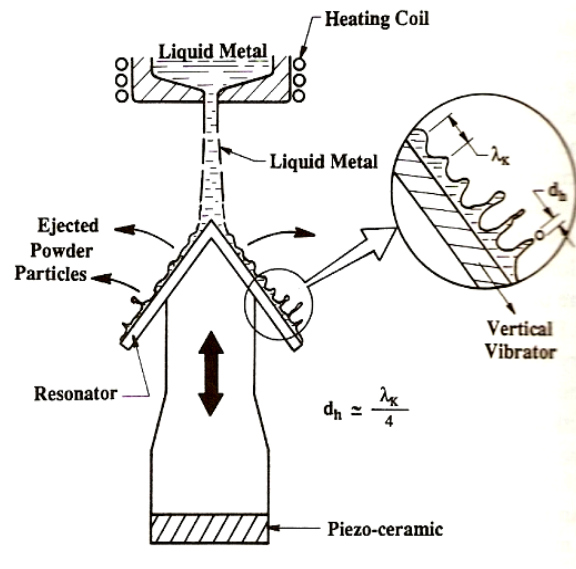


$$d_{50} = \sqrt{\frac{A \cdot \sigma}{\rho_m \omega^2 R}}$$

σ is the melt surface tension
 A is the constant
 ρ_m is the melt density
 ω is the angular speed
 R is the disc radius

Ultrasonic Atomization

Ultrasonic atomization is a technique used to produce a fine spray from a liquid. Procedure requires the use to ultrasonic wave frequency for liquid disintegration into a fine particle



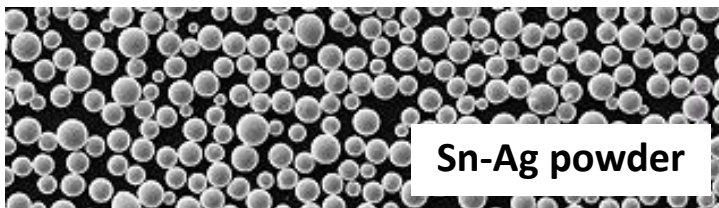
Mechanisms of melt disintegration by UA

$$D_p = 0.34 \left(\frac{8\pi\sigma}{\rho f^2} \right)^{1/3}$$

σ is the melt surface tension

ρ is the melt density

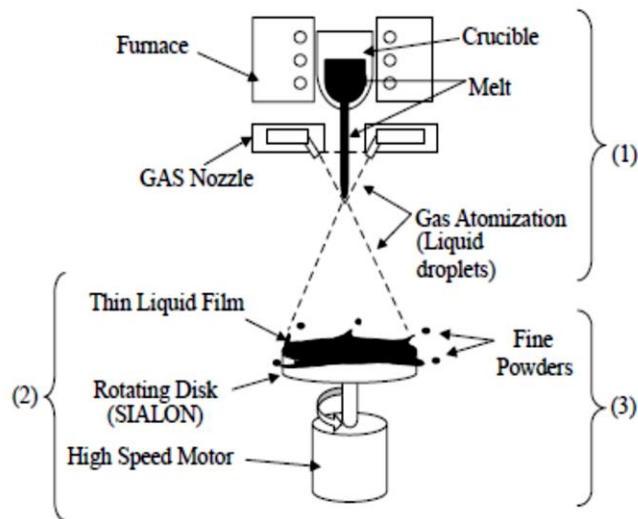
f is the ultrasonic frequency



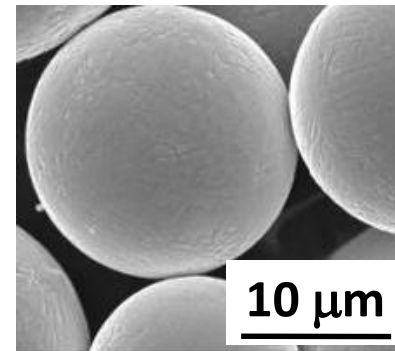
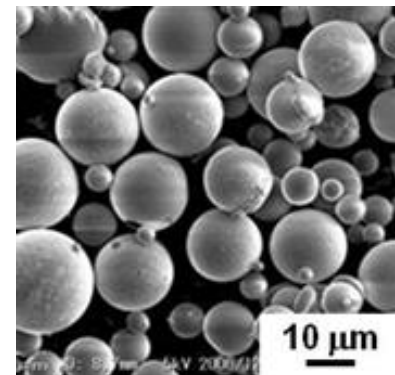
Sn-Ag powder

Hybrid Atomization

Hybrid atomization is a new atomization technique that combines gas atomization with centrifugal atomization. This process can produce fine, spherical powders economically with a mean size of about 10 μm diameter.



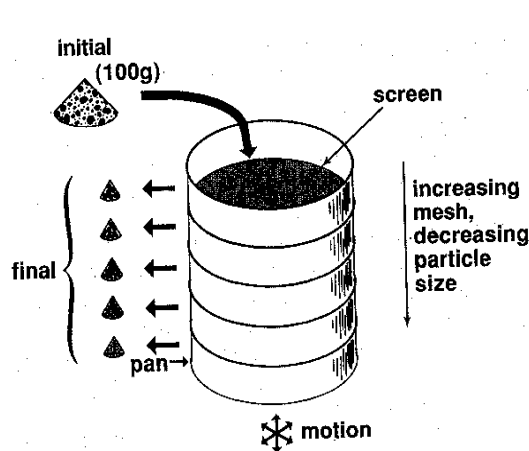
- (1) Gas atomization
- (2) Thin liquid film formation on disk
- (3) Centrifugal breakup and powder formation



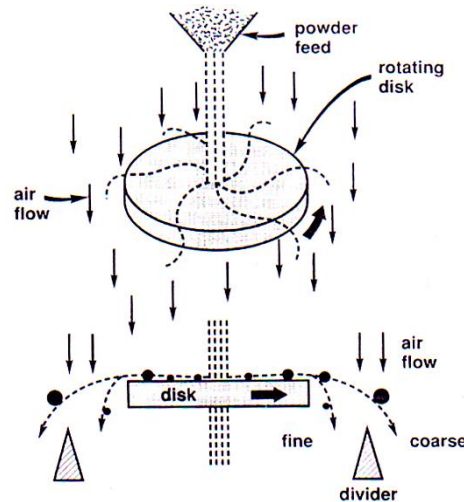
Lead-free solder powder

Powder Characterization

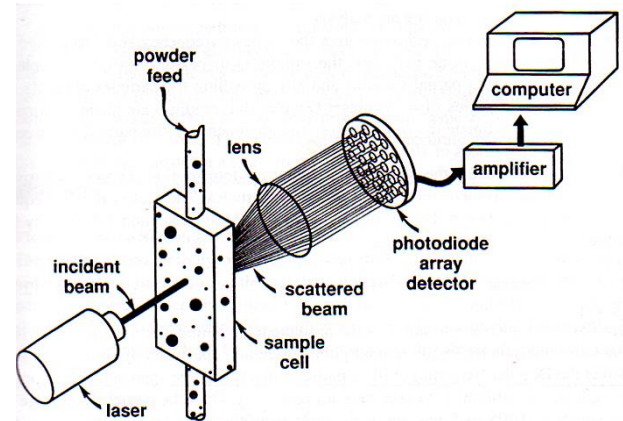
Particle size: determination of the dimensions of a particle. The size of a particle depends on the measurement technique and particle shape.



Sieve analysis
(ref. R.M. German)



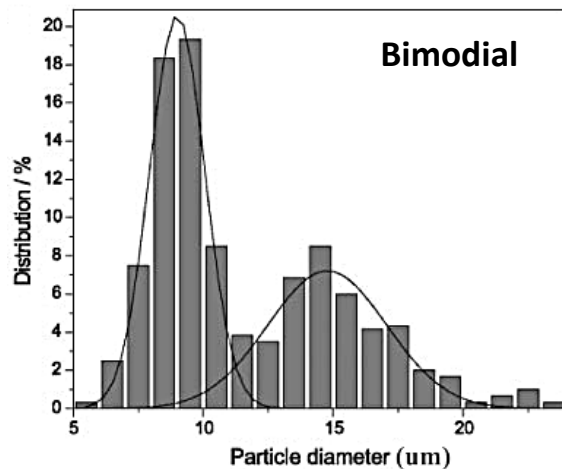
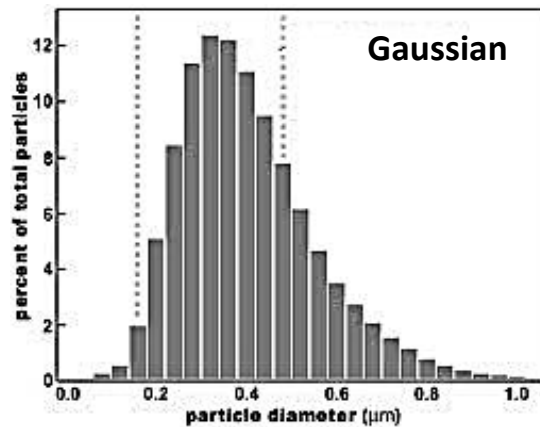
**Air classification
using a rotating
disc**
(ref. R.M. German)



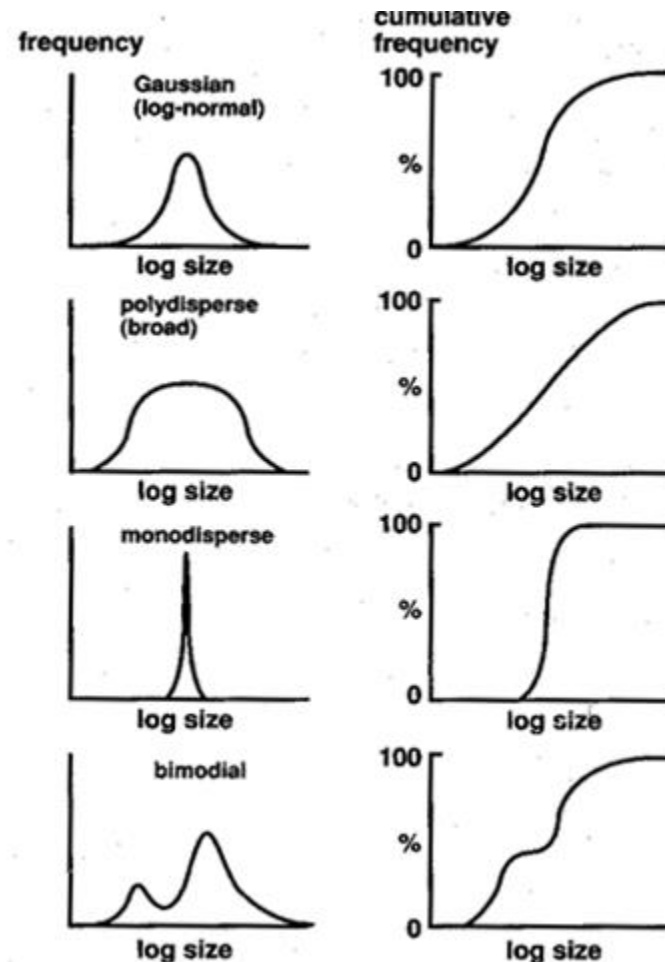
**Laser light
scattering**
(ref. R.M. German)

Particle size data

Particle size distribution



Common types of size distribution

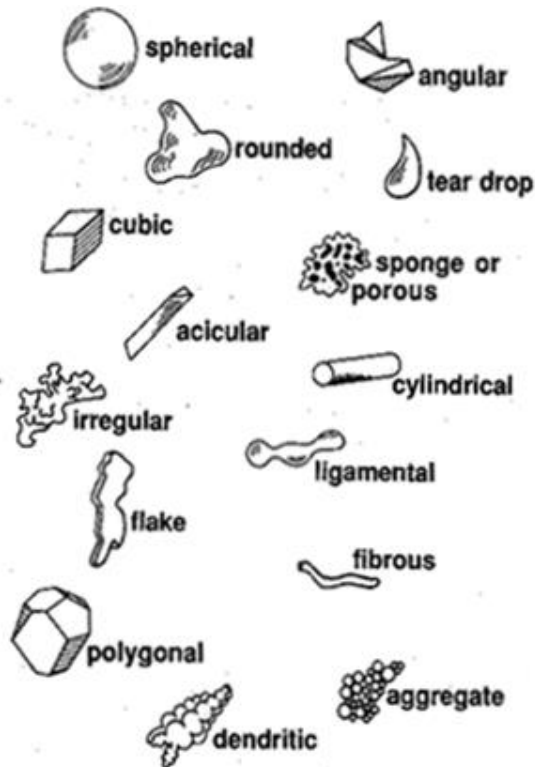


(ref. R.M. German)

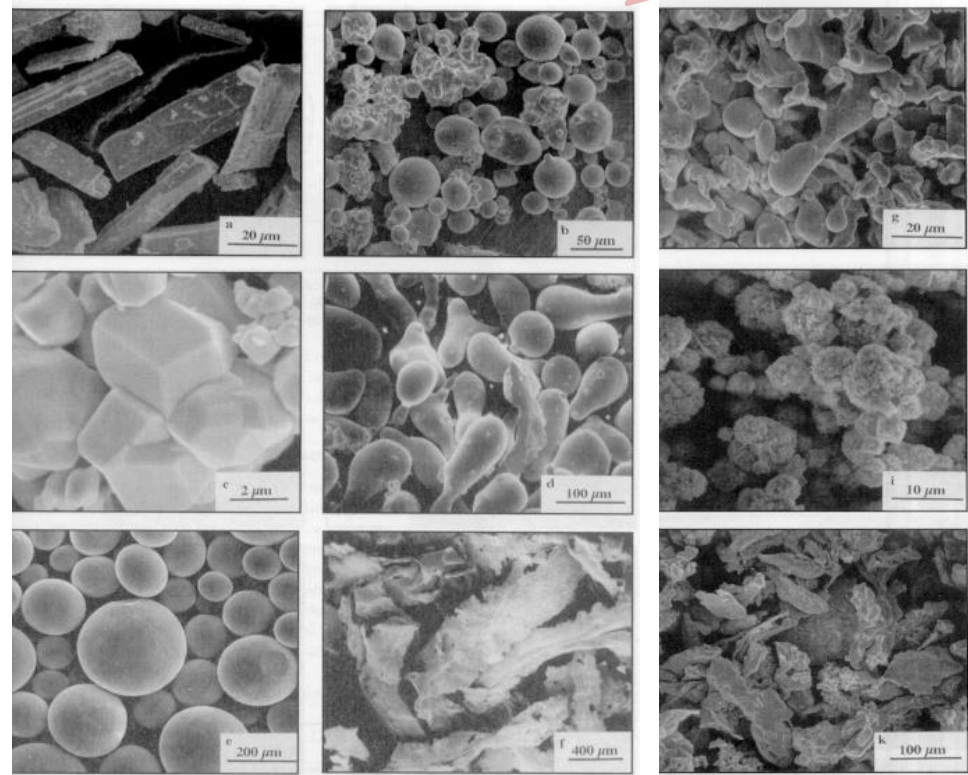
Particle shape

Particle shape influences packing, flow, and compressibility. Particle shape varies with size and manufacturing technique.

OM & SEM



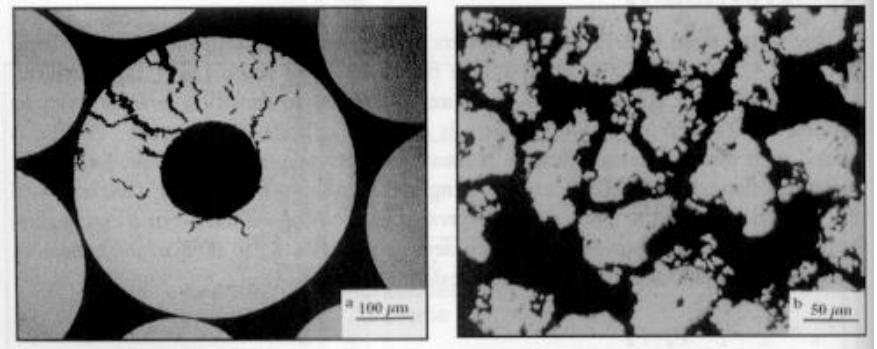
(ref. R.M. German)



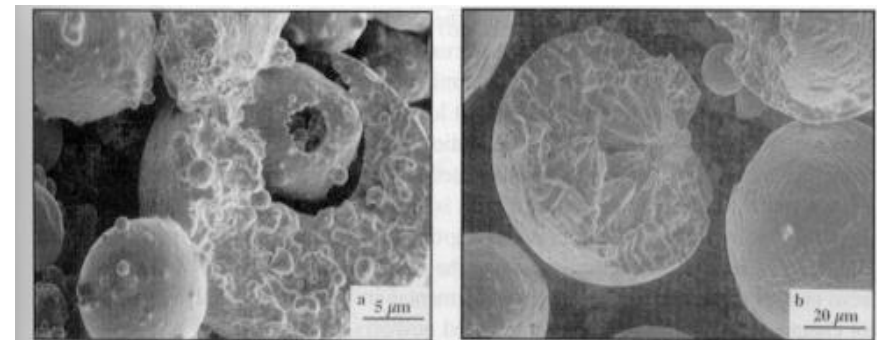
Powder Structure

A magnified cross section through a powder reveals artifacts that indicate

- Fabrication conditions
- Possible processing problems
- Internal pores, segregation, inclusions, and oxides
- Fractured



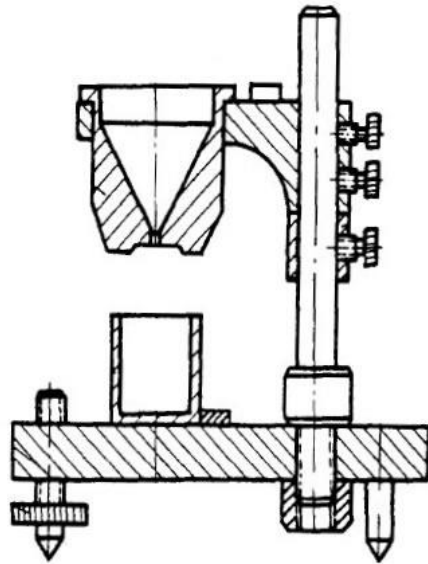
Cross-section metal powder internal details (photo courtesy of J. Dunlap)



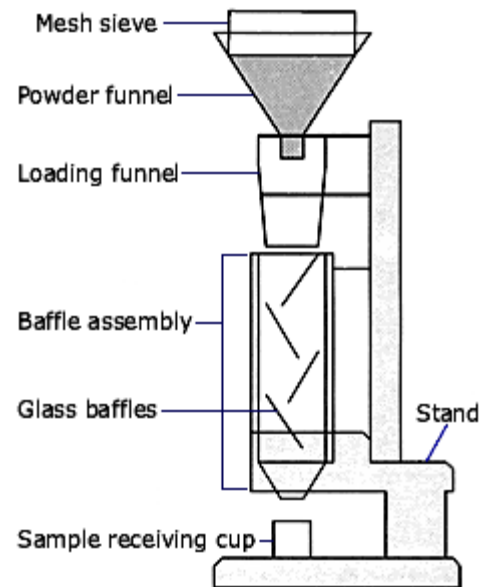
Fractured centrifugally atomized particle (photo courtesy of J. Rawers)

Powder density and flow

The basic components of the Hall flowmeter and Scott volumeter for measuring the flow and density of powders



Hall flowmeter
(ref. R.M. German)

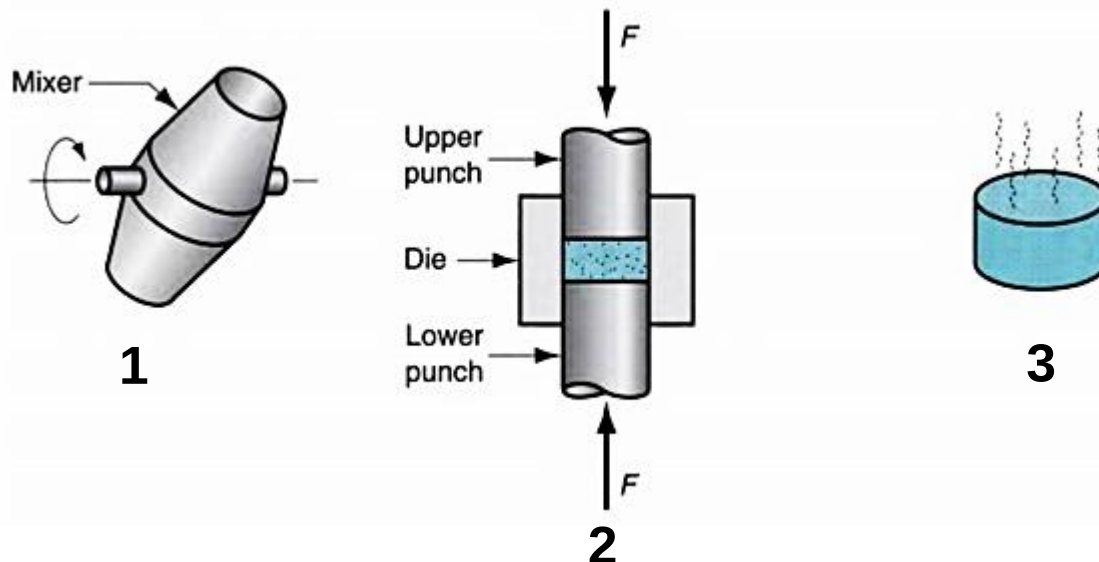


Scott volumeter
(ref. R.M. German)

Powder compaction and sintering

■ Conventional press and sinter steps

1. Blending and mixing of powders
2. Compaction - pressing into desired shape
3. Sintering - heating to temperature below melting point to cause solid-state bonding of particles and strengthening of part



Powder compaction and sintering

❑ Blending and mixing of powders

The starting powders must be homogenized

- **Blending** - powders of the same chemistry but possibly different particle sizes are intermingled
 - Different particle sizes are often blended to reduce porosity
- **Mixing** - powders of different chemistries are combined

Link VDO: <https://www.youtube.com/watch?v=BixMInFlb1I>

Powder compaction and sintering

□ Compaction

High pressure to form the powders into the required shape

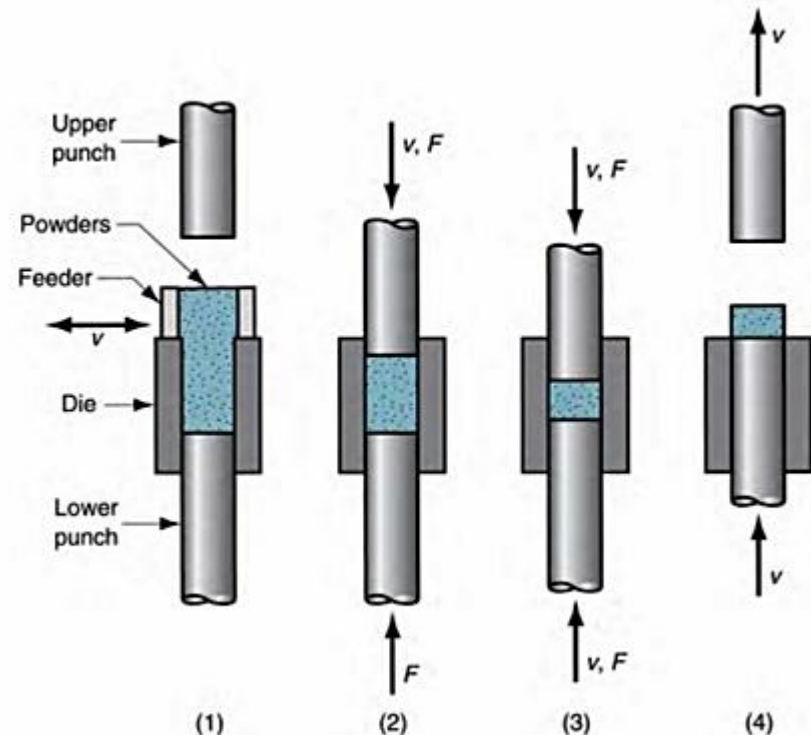
- Conventional compaction method is pressing, in which opposing punches squeeze the powders contained in a die
- The workpart after pressing is called a **green compact**
- The green strength of the part should be adequate for handling

Powder compaction and sintering

❑ Conventional pressing in PM

Pressing in PM:

- (1) filling die cavity with powder by automatic feeder
- (2) Initial
- (3) final positions of upper and lower punches during pressing
- (4) part ejection

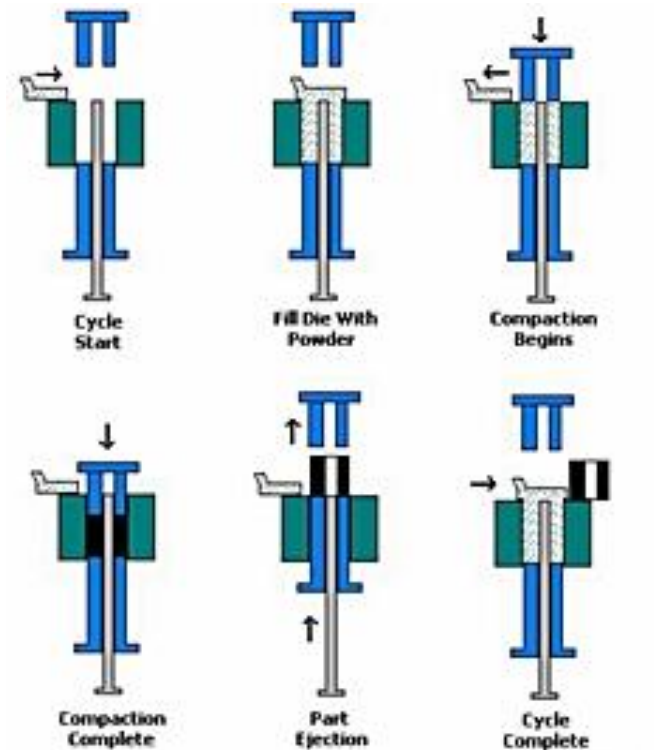
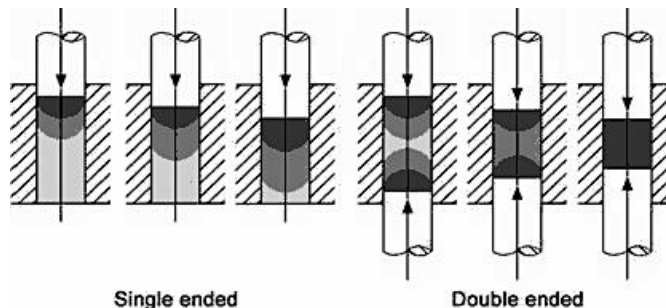


Link VDO: <https://www.youtube.com/watch?v=5VmelunoyKw>

Powder compaction and sintering

□ Die compaction technique

- Pressure applied mechanically or hydraulically
- Single acting compaction
 - Thin parts like washers, discs, thin ring etc.
- Double acting compaction
 - Bushings, bearings etc.
- Double acting floating die compaction



Ref. http://www.keramverband.de/brevier_engl/4/1/4_1_3.htm

Powder compaction and sintering

❑ Conventional pressing in PM

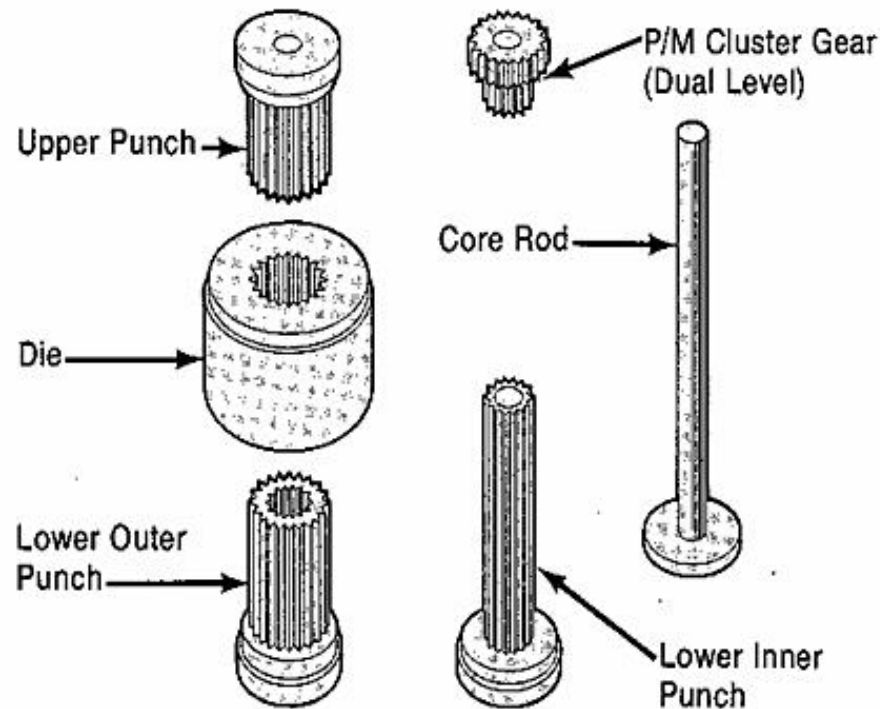


Figure 4 P/M Tooling For A Two-Level Part

Powder compaction and sintering

- ❑ Loose powder is compacted and densified into a shape, known as green compact
- ❑ Most compacting is done with mechanical presses and rigid tools
 - Hydraulic and pneumatic presses are also used

TABLE 18-1 Typical Compacting Pressures for Various Applications

Application	Compaction Pressures	
	tons/in. ²	Mpa
Porous metals and filters	3–5	40–70
Refractory metals and carbides	5–15	70–200
Porous bearings	10–25	146–350
Machine parts (medium-density iron & steel)	20–50	275–690
High-density copper and aluminum parts	18–20	250–275
High-density iron and steel parts	50–120	690–1650

Powder compaction and sintering

□ Density lines after compacted powder

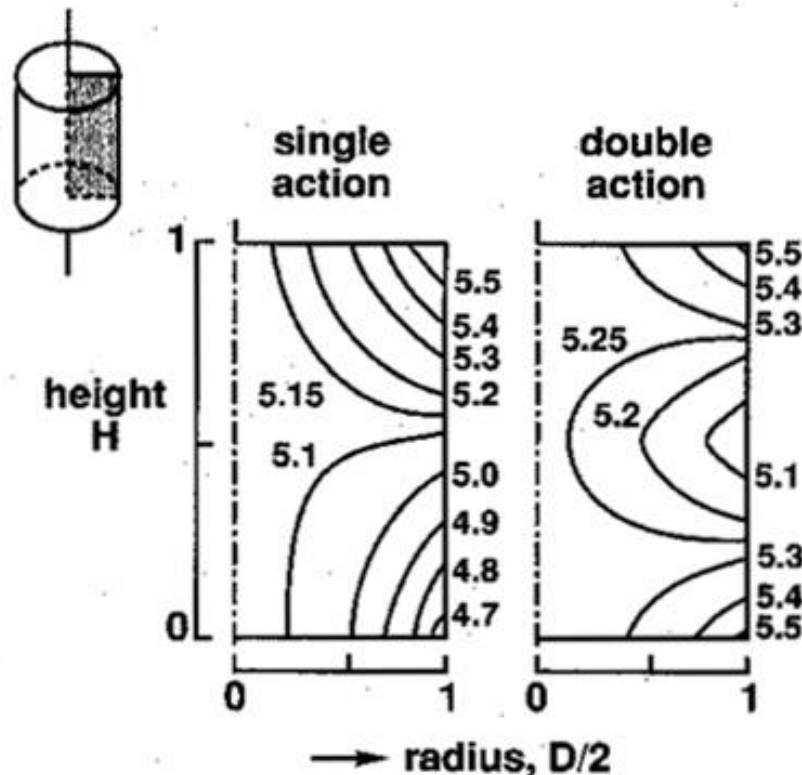


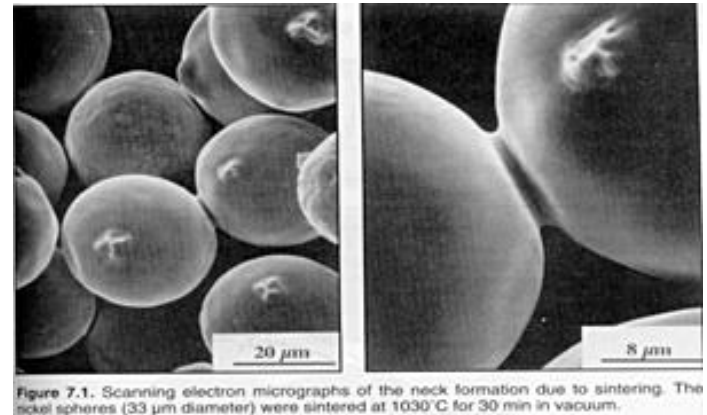
Figure 6.32. Constant density lines in cylinders of compacted copper powder. The contours are in g/cm^3 and are plotted versus height and radius (from the centerline axis) for both single- and double- action compaction.

Powder compaction and sintering

□ Sintering

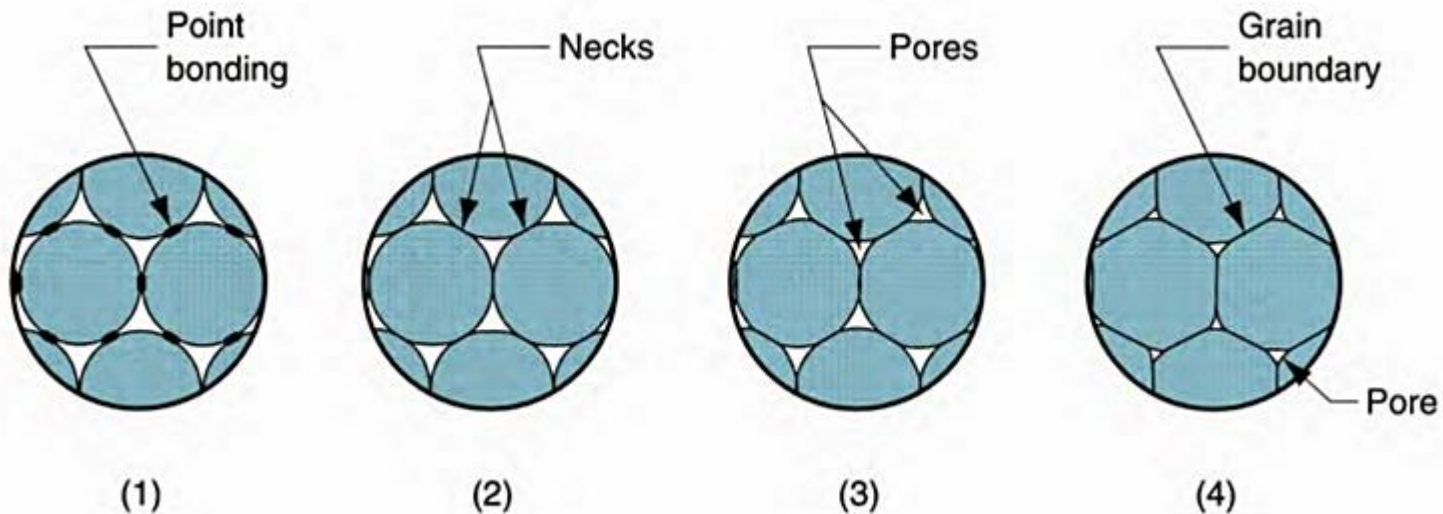
Heat treatment to **bond** the metallic particles, thereby increasing strength and hardness

- Usually carried out at 70% to 90% of the metal's melting point (absolute scale)
- The primary driving force for sintering is reduction of surface energy
- Part shrinkage occurs during sintering due to pore size reduction



Powder compaction and sintering

□ Sintering sequence on a microscopic scale



(1) Particle bonding is initiated at contact points; **(2)** contact points grow into "necks"; **(3)** pores between particles are reduced in size; **(4)** grain boundaries develop between particles in place of necked regions

Powder compaction and sintering

□ Pore structures in sintering

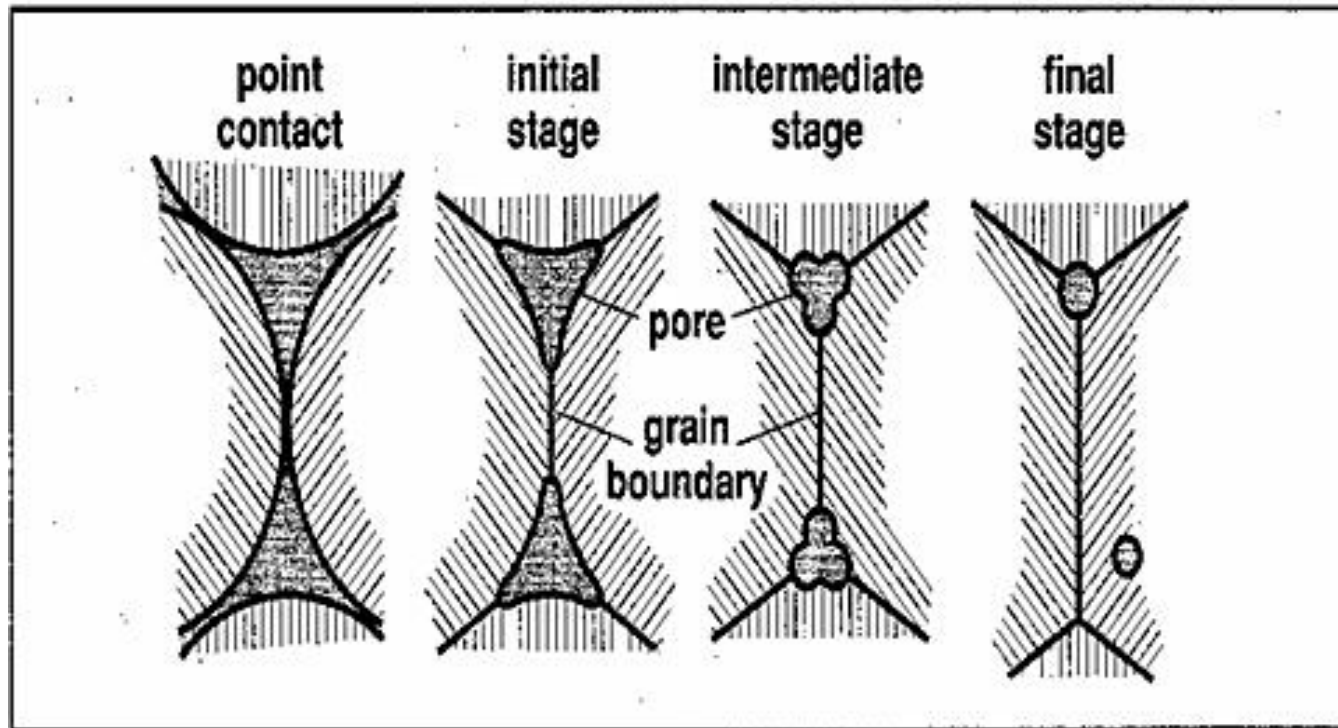


Figure 7.17. A schematic diagram of the pore structure changes during sintering, starting with particles in point contact. The pore volume decreases and the pores become smoother. As pore spheroidization occurs, the pores are replaced by grain boundaries.

Powder compaction and sintering

❑ Liquid phase sintering

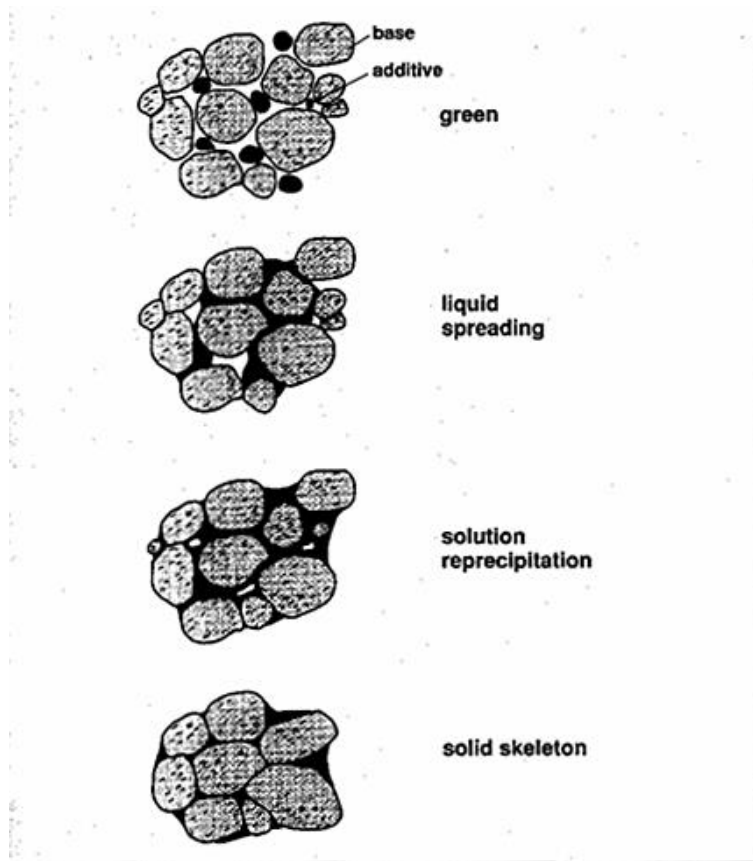
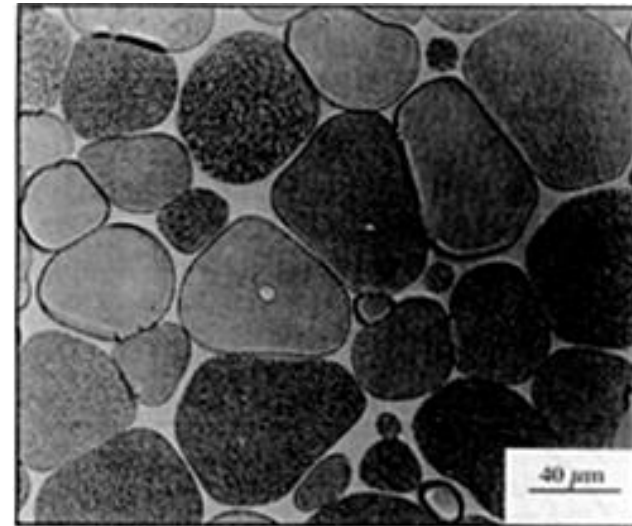


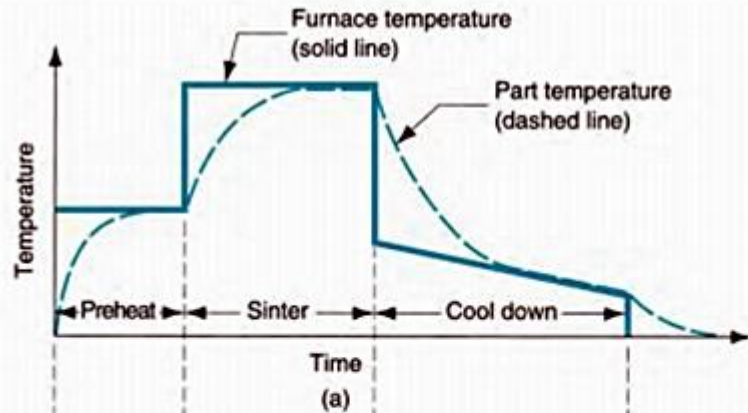
Figure 7.31. The conceptual stages to liquid phase sintering using a mixture of two powders. The base powder remains solid during the process and the additive powder is possible for formation of the liquid.



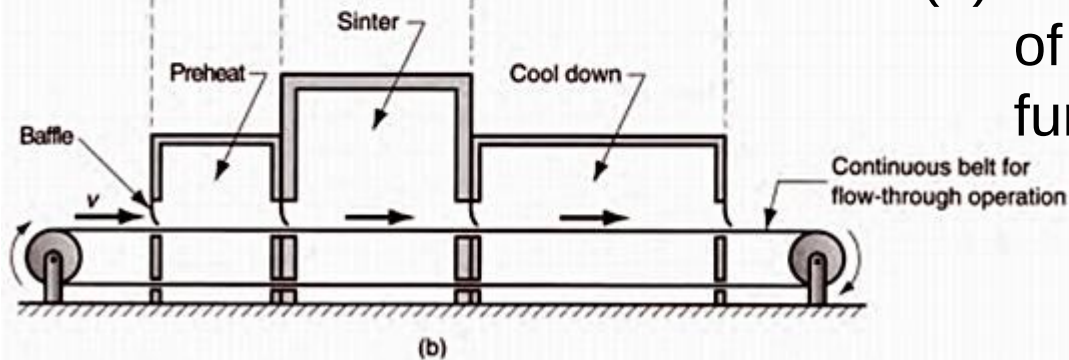
The microstructure of a liquid phase sintered heavy alloy of 95%W-3.5%Ni-1.5Fe sintered at 1475 °C. The tungsten grains have coarsened considerably from the original 3 μm particle size and the grain shape has accommodated a high packing density

Powder compaction and sintering

□ Sintering cycle and furnace

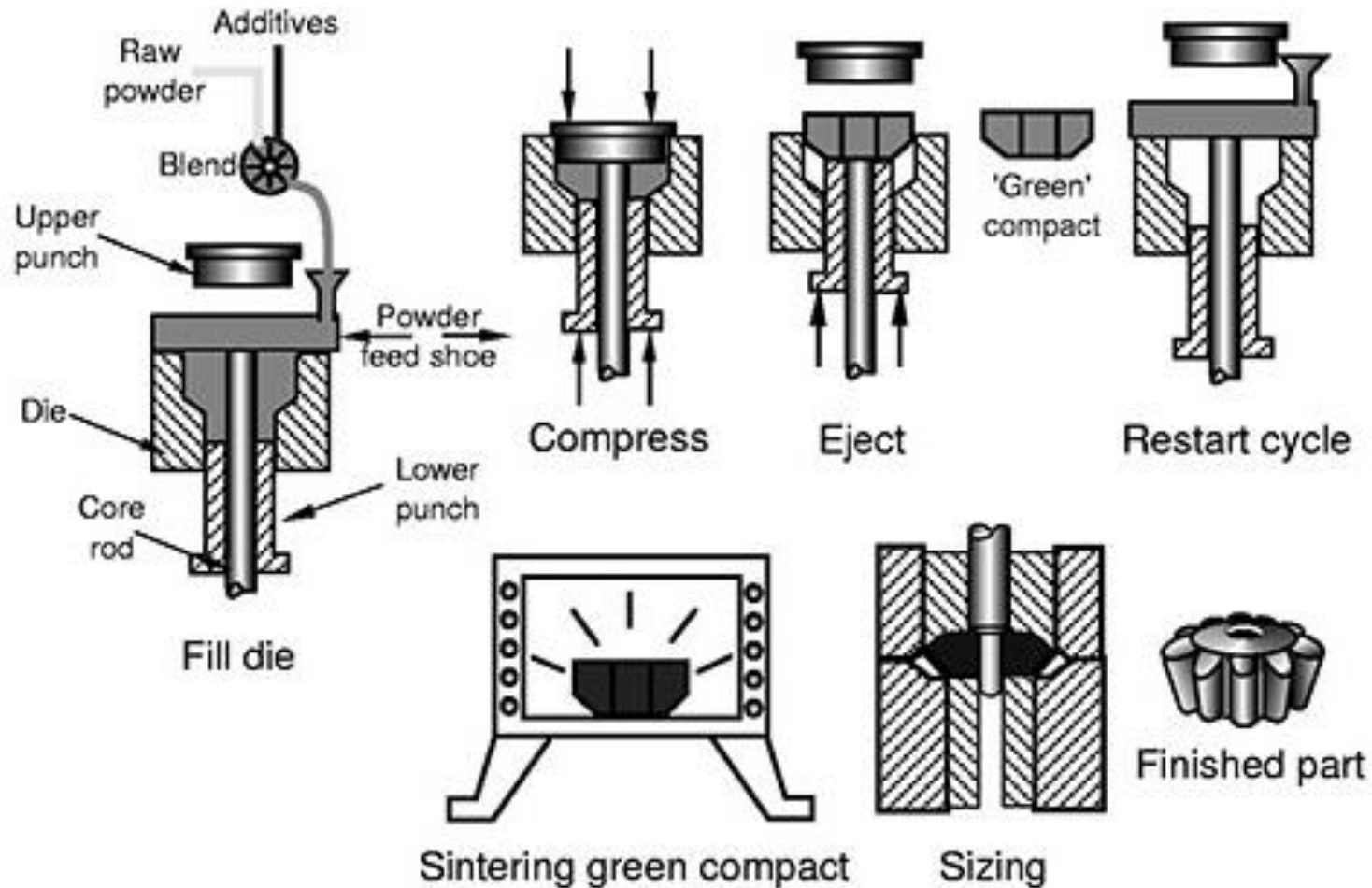


(a) Typical heat treatment cycle in sintering



(b) Schematic cross section of a continuous sintering furnace

Powder compaction and sintering



Method of part production

Powder compaction and sintering

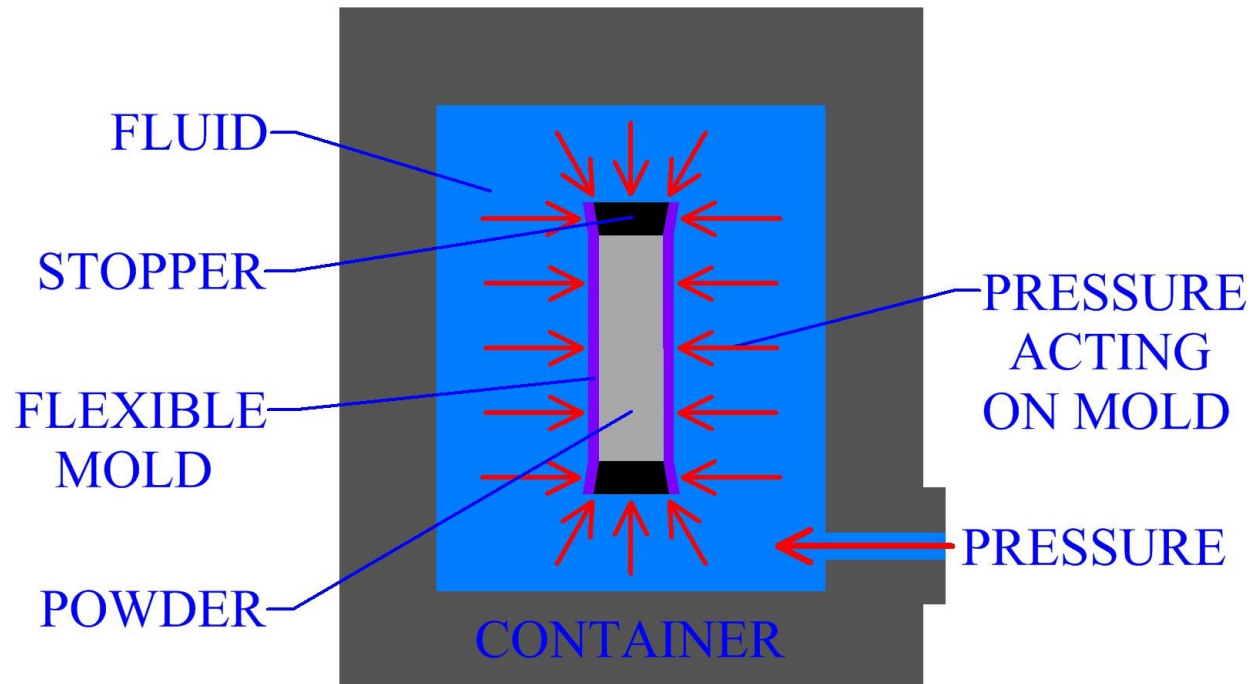
❑ Alternative pressing

Some additional methods for producing PM parts:

- **Isostatic pressing** - hydraulic pressure is applied from all directions to achieve compaction
- **Powder injection molding (PIM)** - starting polymer has 50% to 85% powder content
 - Polymer is removed and PM part is sintered
- **Hot pressing** - combined pressing and sintering

Powder compaction and sintering

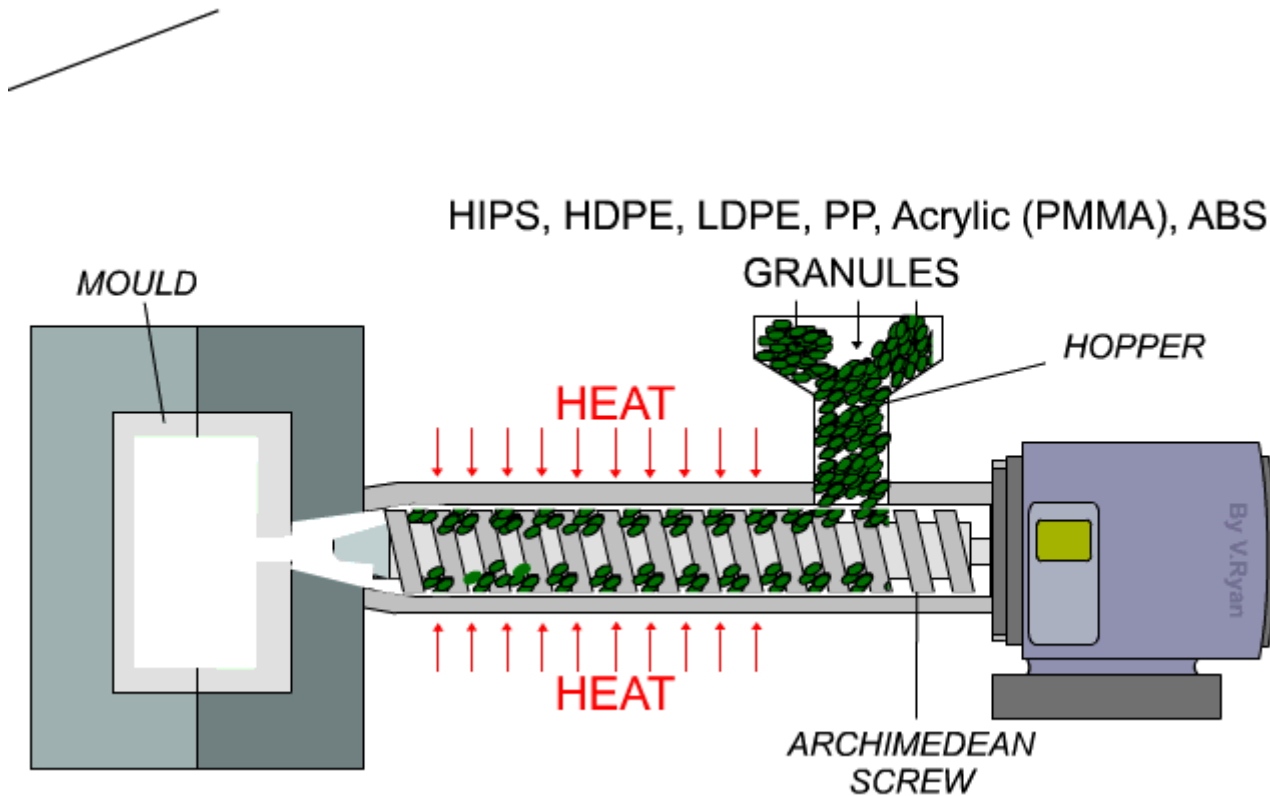
□ Isostatic pressing



Ref. http://thelibraryofmanufacturing.com/alt_powder_processes.html

Powder compaction and sintering

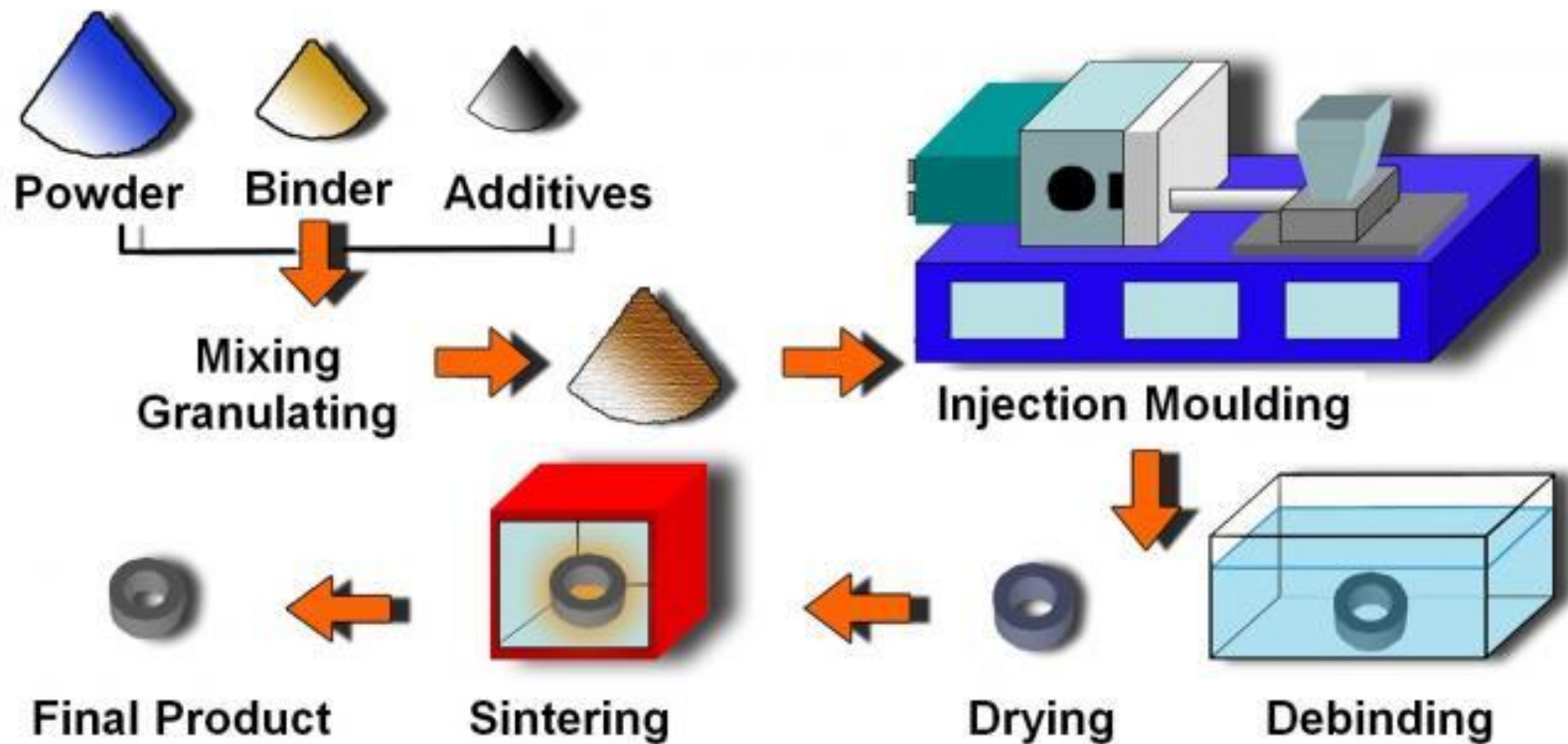
❑ Powder injection molding (PIM)



Ref. <http://www.che.hw.ac.uk/teaching/B11MS1/Material/Processing/Proc1.htm>

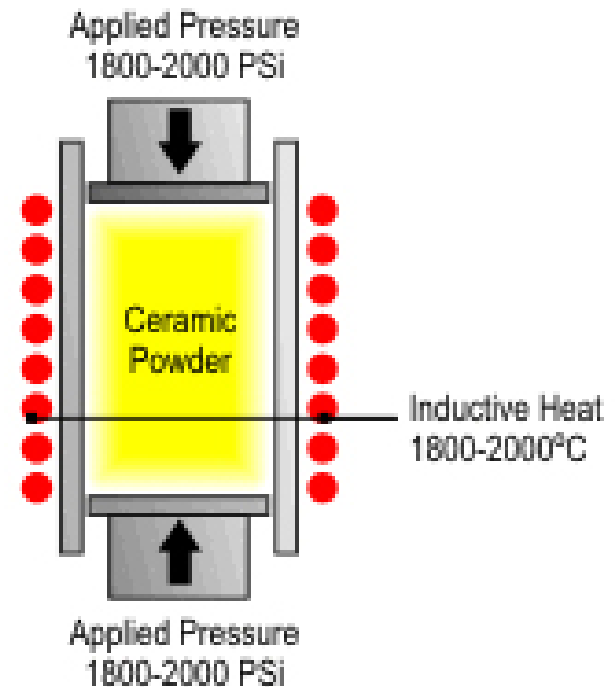
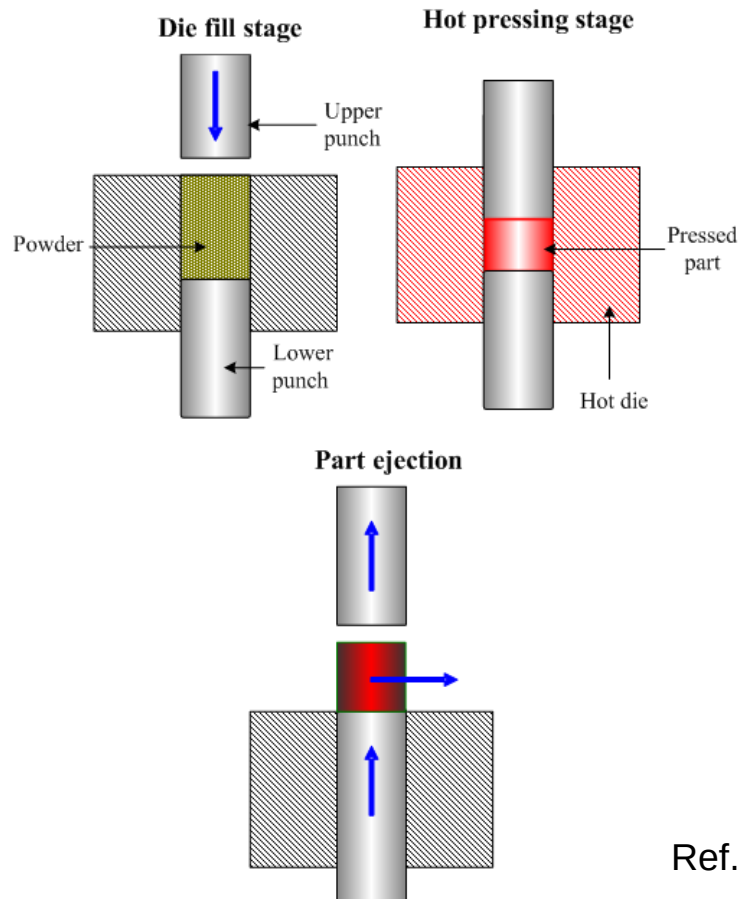
Powder compaction and sintering

□ Powder injection molding (PIM)



Powder compaction and sintering

□ Hot pressing



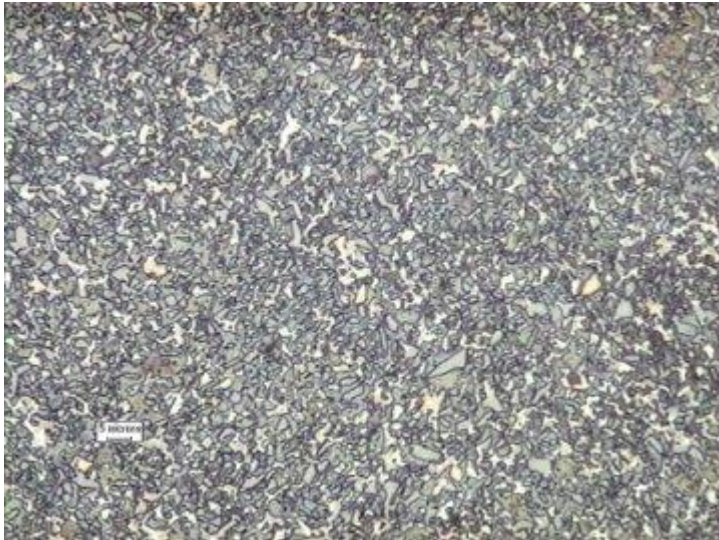
Ref. <http://www.dynacer.com/processing/hot-pressing/>

Powder applications

❑ Powder metallurgy

Cermet cutting tools
(Ceramic-Metal composite)

Microstructure: ceramic particles in metal matrix



Ref. www.moldmakingtechnology.com

Cermet cutting inserts for lathe



Ref. www.kyocera-yashica.com.br

Cermet-tipped saw blade for long life



Ref. www.toolbarn.com

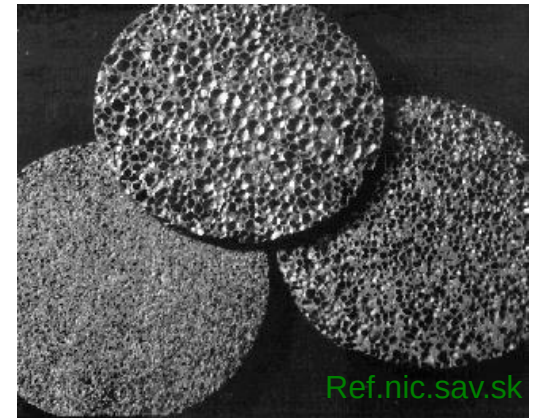
Powder applications

□ Porous metals

Oil-impregnated Porous Bronze Bearings



Metal filters



Ref. www.ondrives.com

Powder applications

❑ Connecting rods



Forged on left; P/M on right

Powder applications

❑ Powdered metal transmission gear



- Warm compaction method with 1650-ton press
- Teeth are molded net shape: No machining
- UTS = 155,000 psi
- 30% cost savings over the original forged part

References

1. **German, R. M. (1994). Powder Metallurgy Science, 2nd Ed, MPIF, USA.**
2. **Powder Metallurgy, ASM Handbook vol. 7, E. Klar (editor), 9th ed., ASM International, Ohio, 1984.**